

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12408313
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Iwasaki; Masataka

System for preparing component feed unit

Abstract

A component supply unit set-up system includes a container warehouse configured to store multiple component containers for containing multiple components, a unit storage shed configured to store multiple component supply units used when the component, which is contained in the component container, is supplied in a mounting operation performed by a component mounter, a work table disposed between the container warehouse and the unit storage shed and used for a loading operation of loading the component supply unit with the component container, a container conveyance robot configured to convey the component container between the container warehouse and the work table, and a unit conveyance robot configured to convey the component supply unit between the unit storage shed and the work table.

Inventors:	Iwasaki; Masataka (Anjo, JP)
Applicant:	FUJI CORPORATION (Chiryu, JP)
Family ID:	76864012
Assignee:	FUJI CORPORATION (Chiryu, JP)
Appl. No.:	17/791420
Filed (or PCT Filed):	January 14, 2020
PCT No.:	PCT/JP2020/000931
PCT Pub. No.:	WO2021/144864
PCT Pub. Date:	July 22, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20230035634 A1	Feb. 02, 2023

Publication Classification

Int. Cl.: H05K13/08 (20060101); H05K13/02 (20060101)

U.S. Cl.:

CPC H05K13/086 (20180801); H05K13/021 (20130101); Y10T29/53178 (20150115)

Field of Classification Search

CPC: H05K (13/086); H05K (13/02); H05K (13/021); Y10T (29/53187); Y10T (29/53417); Y10T (29/53178)

USPC: 29/740; 29/739; 29/742; 29/832

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5503516	12/1995	Kasai	414/800	B23P 19/001
10820459	12/2019	Kondo	N/A	H05K 13/08
2018/0220560	12/2017	Otsuka et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
20 2016 103 393	12/2016	DE	N/A
2011-3801	12/2010	JP	N/A
WO 2014/010083	12/2013	JP	N/A
2014-197594	12/2013	JP	N/A
2018-125383	12/2017	JP	N/A
WO 2018/173204	12/2017	JP	N/A
2019-91771	12/2018	JP	N/A
WO 2019/142336	12/2018	JP	N/A

OTHER PUBLICATIONS

International Search Report issued Mar. 24, 2020 in PCT/JP2020/000931, filed on Jan. 14, 2020, 3 pages. cited by applicant

Primary Examiner: Phan; Thiem D

Attorney, Agent or Firm: Oblon, McClelland, Maier & Neustadt, L.L.P.

Background/Summary

TECHNICAL FIELD

(1) The present specification relates to a set-up system corresponding to a loading operation for loading a component supply unit with a component container and related operations before and after the loading operation.

BACKGROUND ART

(2) A technique of mass-producing board products by performing board work on a board on which printed wiring is performed has become widespread. Further, it is common to provide multiple types of board work machines for performing the board work side by side so as to configure a board work line. A component mounter among the board work machines uses a component supply unit loaded with the component container that contains multiple components. The loading operation for loading the component supply unit with the component container, in other words, a set-up operation for preparing the component supply unit for use, is often performed in a set-up area distant from the component mounter being operated (external set-up).

(3) This loading operation has been conventionally performed by human hand and requires a lot of labor. Further, the loading operation is performed ahead of schedule to ensure that the production plan is achieved, which increases the number of work-in-process component supply units that cannot be diverted to other uses. Therefore, the management is complicated and further labor is required. In recent years, for the purpose of labor saving, a container loader that automates at least a part of the loading operation has been put into practical use. One example of a technique related to the automation of the loading operation is disclosed in Patent Literature 1.

(4) A second embodiment of Patent Literature 1 describes a configuration including a storage shed for storing a reel (one example of the component container), a storage shed for storing a feeder (one example of the component supply unit), a reel set device (one example of the container loader), a conveyance device, and a management section. The reel set device performs the loading operation for setting a reel to a feeder. The conveyance device conveys the set-up feeder to the component mounter. The management section controls the delivery, the loading operation, the conveyance operation, or the like of the reels and the feeders, and manages the production of the component mounter. According to this configuration, it is said that the combination and the location of the reel and the feeder can be easily recognized so that the reel and the feeder can be easily managed.

PATENT LITERATURE

(5) Patent Literature 1: International Publication No. 2019/142336

BRIEF SUMMARY

Technical Problem

(6) Incidentally, in the reel set device of Patent Literature 1, it is preferable that the loading operation itself is automated to achieve labor saving. However, the operation of conveying the reel and the feeder from the storage shed to the reel set device or the operation of loading the set-up feeder on the conveyance device is manually performed. In other words, the related operations before and after the loading operation still require human hands.

(7) An object of the present specification is to provide a component supply unit set-up system that achieves labor saving in related operations before and after a loading operation for loading a component supply unit with a component container.

Solution to Problem

(8) The present specification discloses a component supply unit set-up system including a container warehouse configured to store multiple component containers for containing multiple components, a unit storage shed configured to store multiple component supply units used when the components, which are contained in the component containers, are supplied in a mounting operation performed by a component mounter, an operation table disposed between the container warehouse and the unit storage shed and used for a loading operation to load the component supply unit with the component container, a container conveyance robot configured to convey the component container between the container warehouse and the operation table, and a unit conveyance robot configured to convey the component supply unit between the unit storage shed and the operation table.

Advantageous Effects

(9) In a component supply unit set-up system of the present specification, a component container is

conveyed from a container warehouse to an operation table by a container conveyance robot, and a component supply unit is conveyed from a unit storage shed to the operation table by a unit conveyance robot. Further, the component supply unit, which is loaded with the component container on the operation table in a loading operation is returned to the unit storage shed by the unit conveyance robot. Therefore, labor saving of related operations such as conveyance performed before and after the loading operation is achieved regardless of whether the loading operation is automated.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a perspective view schematically illustrating a configuration of a component supply unit set-up system of a first embodiment.
- (2) FIG. 2 is a front view schematically illustrating a configuration of the component supply unit set-up system of the first embodiment.
- (3) FIG. 3 is a functional block diagram illustrating a configuration of control in the first embodiment.
- (4) FIG. 4 is a diagram illustrating an operation flow for describing a control operation of a set-up control section in the first embodiment.
- (5) FIG. 5 is a perspective view schematically illustrating a component supply unit set-up system of a second embodiment.
- (6) FIG. 6 is a perspective view schematically illustrating a part of a component supply unit set-up system of a third embodiment.
- (7) FIG. 7 is a plan view schematically illustrating the component supply unit set-up system of the third embodiment.

DESCRIPTION OF EMBODIMENTS

1. Configuration of Component Supply Unit Set-Up System 1 of First Embodiment

- (8) A configuration of component supply unit set-up system 1 of a first embodiment will be described with reference to the configuration views of FIGS. 1 and 2, and the functional block diagram of FIG. 3. Set-up system 1 corresponds to a loading operation for loading component supply unit 92 with component container 91 and the related operations before and after the loading operation. Set-up system 1 includes container warehouse 2, unit storage shed 3, operation table 4, container conveyance robot 5, unit conveyance robot 6, set-up control section 8, and the like.
- (9) Component container 91 contains multiple components. Component supply unit 92 is used when supplying the components contained in component container 91 in a mounting operation performed by component mounter 98. Identification codes indicating each type or individual are attached to component container 91 and component supply unit 92. The identification code of component container 91 includes information indicating the type of the component contained in component container 91. As the identification code, a barcode, a two-dimensional code, or the like is used.
- (10) As component container 91, a tape reel can be exemplified, and as component supply unit 92, an integral type feeder device in which the tape reel is directly loaded can be exemplified. A carrier tape in which multiple components are sealed at a predetermined pitch is wound and held on the tape reel. The feeder device is installed on component mounter 98 in a form in which the tape reel is loaded. The feeder device supplies the component to a component mounting tool of component mounter 98 by pulling out the carrier tape from the tape reel and sending the carrier tape to a component pick up position. When the mounting operation of the component proceeds and the component of the tape reel is consumed, the entire feeder device is exchanged.
- (11) The feeder device is not limited to an integral type and may be a separate type. The separate

type feeder device is configured to include a feeder main body portion having a feeding mechanism of the carrier tape and a reel cassette in which the tape reel is loaded. The feeder main body portion is permanently provided in component mounter **98**, and the separate reel cassette is disposed in the vicinity of the feeder main body portion. When the mounting operation of the component proceeds and the component of the tape reel is consumed, the reel cassette is exchanged. The reel cassette corresponds to component supply unit **92** in which component container **91** is loaded.

(12) Hereinafter, a case where component supply unit **92** is an integral type feeder device and component container **91** is a tape reel will be described. A tray may be used as component container **91**, and a tray loading unit may be used as component supply unit **92**.

(13) Container warehouse **2** stores multiple component containers **91**. As illustrated in FIGS. **1** and **2**, container warehouse **2** is formed in a large vertical substantially octagonal prism shape.

Container warehouse **2** has a storage port on the rear side and retrieval port **21** at a substantially mid-height of the left side face. As illustrated in FIG. **3**, multiple storing positions, actuator **22**, and code reader **23** are provided in container warehouse **2**. Actuator **22** transfers component container **91** stored in the storage port to the storing position and transfers component container **91** in the storing position to retrieval port **21**. Actuator **22** is controlled by warehouse control section **29**.

(14) Code reader **23** reads the identification code attached to stored component container **91** and delivers the read result to warehouse control section **29**. Therefore, warehouse control section **29** can recognize the individuals of all stored component containers **91** in association with the storing positions. As illustrated in FIGS. **1** and **2**, warning lamp **24** is provided on the left front side of the upper portion of container warehouse **2**. Warehouse control section **29** controls the lighting of warning lamp **24** when an abnormality occurs or a failure occurs and notifies an operator of the same.

(15) Unit storage shed **3** stores multiple component supply units **92**. Unit storage shed **3** includes base **31** and storage rack **35**. Base **31** is formed in a rectangular parallelepiped frame shape. Upper section guide **32** extending in the left-right direction is provided on an upper portion of the front face of base **31**. Lower section guide **33** extending in the left-right direction is provided on a lower portion of the front face of base **31**. Upper section guide **32** and lower section guide **33** are formed in, for example, a protrusion shape protruding ahead or a groove shape opened upward.

(16) Storage rack **35** is formed in a box shape opening on the front side and is fixed to an upper portion of base **31**. Multiple storage slots arranged in the left-right direction are formed on a bottom face of the inner side of storage rack **35**. Storage slot is formed with, for example, a groove extending in the front-rear direction, into which component supply unit **92** is inserted from the front side and stored. Component supply unit **92** stored in unit storage shed **3** does not matter whether component container **91** is loaded. In FIGS. **1** and **2**, one component supply unit **92** is illustrated, and in actuality, multiple component supply units **92** are stored side by side in the left-right direction.

(17) Multiple storing side connectors **36**, which are arranged in the left-right direction corresponding to the storage slots, are provided on a rear face of the inner side of storage rack **35**. When component supply unit **92** is stored in the storage slot, the unit side connector provided on the rear side of component supply unit **92** is automatically fitted into storing side connector **36**. As a result of this fitting, component supply unit **92** is supplied with power and is communicatively connected to storage shed control section **39** (see FIG. **3**). Component supply unit **92** stores the information on its own identification code in a built-in memory. Therefore, storage shed control section **39** can acquire the identification codes of all stored component supply units **92** by the communication and recognize the identification code in association with the position of the storage slot.

(18) Unit storage shed **3** is capable of being additionally installed. When additional unit storage shed **3** is installed, multiple unit storage sheds **3** are disposed adjacent to each other in the left-right direction. As a result, multiple upper section guides **32** are connected in the left-right direction to

form a long movement guide. Similarly, multiple lower section guides **33** are connected in the left-right direction to form a long movement guide.

(19) Operation table **4** is disposed between container warehouse **2** and unit storage shed **3**. Operation table **4** is used for the loading operation to load component supply unit **92** with component container **91**. Operation table **4** includes base **41** and operation jig **44**. Base **41** is formed in a rectangular parallelepiped frame shape having a height equivalent to that of base **31** of unit storage shed **3**. Upper section guide **42** extending in the left-right direction is provided on an upper portion of the front face of base **41**. Lower section guide **43** extending in the left-right direction is provided on a lower portion of the front face of base **41**.

(20) Upper section guide **42** and lower section guide **43** are disposed at the same height as upper section guide **32** and lower section guide **33** of unit storage shed **3** and are formed in the same shape. Upper section guide **42** and upper section guide **32** are connected side by side in the left-right direction to form a long movement guide. Similarly, lower section guide **43** and lower section guide **33** are connected side by side in the left-right direction to form a long movement guide.

(21) Operation jig **44** is attached to an upper portion of base **41** in an attachable and detachable manner. Operation jig **44** has the function of holding component supply unit **92** and facilitates the loading operation (set-up operation) of component container **91** by the operator. Examples of the loading operation include an operation of assembling the tape reel corresponding to component container **91** with respect to the feeder device corresponding to component supply unit **92**, an operation of pulling out the carrier tape from the tape reel to cut an unnecessary portion of a distal end, and an operation of advancing the distal end of the carrier tape after cutting to a predetermined position in the feeder device.

(22) Further, code reader **45** is included in operation jig **44** (see FIG. **3**). Code reader **45** reads the identification code attached to component container **91** or component supply unit **92**. As a result, load information **V** (see FIG. **3**) associating an individual of component container **91** with an individual of component supply unit **92**, which is loaded with component container **91**, is created. The creation of load information **V** can be viewed as part of the loading operation. Operation jig **44** can also be used for a removing operation for removing component container **91** from component supply unit **92**.

(23) Container conveyance robot **5** conveys component container **91** between container warehouse **2** and operation table **4**. Container conveyance robot **5** is an arm robot in an embodiment of a non-moving robot that does not require movement. Container conveyance robot **5** includes leg portion **51**, main body portion **52**, and arm portion **53**. Leg portion **51** is provided between container warehouse **2** and operation table **4** disposed close to each other. Leg portion **51** is formed in a table shape having a rectangular upper face.

(24) Main body portion **52** is fixed to the center of the upper face of leg portion **51**. Arm portion **53** is provided on an upper portion of main body portion **52**. Arm portion **53** is configured to be articulated so as to be able to freely operate. End effector **54** suitable for component container **91** is provided at a distal end of arm portion **53**. Examples of end effector **54** include a suction cup that picks up component container **91** with negative pressure, fingers that clamp component container **91**, or the like. Container conveyance robot **5** receives component container **91** at retrieval port **21** of container warehouse **2**, conveys component container **91** to the vicinity of operation jig **44**, and delivers component container **91** to the operator. Container conveyance robot **5** is controlled by set-up control section **8**.

(25) Container conveyance robot **5** may have a conveyance function in the opposite direction. Further, container conveyance robot **5** may be a conveyor robot in another embodiment of a non-moving robot. The conveyor robot rotationally rotates an annular conveyance conveyor and conveys component container **91** mounted on the conveyance conveyor from retrieval port **21** to the vicinity of operation jig **44**.

(26) Unit conveyance robot **6** conveys component supply unit **92** between unit storage shed **3** and

operation table **4**. Unit conveyance robot **6** is formed so as to be longitudinally long. Unit conveyance robot **6** includes a guide member and a movement drive section on the rear side. The guide member engages with the movement guide including multiple section guides (**32**, **33**, **42**, **43**). As a result, the entire weight of unit conveyance robot **6** is supported by the movement guide, and the movement direction is predefined. The movement drive section operates, for example, by using a non-contact power supply device or a battery (not illustrated) as a power source. The movement drive section is configured to include, for example, a combination of traveling wheels and a drive motor, or a movement mechanism to which a linear motor is applied. As a result, unit conveyance robot **6** is moved the front side of unit storage shed **3** and operation table **4** along the movement guide.

(27) Unit conveyance robot **6** further includes a unit holding space and a unit operating mechanism. The unit holding space is a space defined in the inner portion of unit conveyance robot **6**. The unit holding space temporarily holds component supply unit **92** to be conveyed. The unit operating mechanism receives and delivers component supply unit **92** between the unit holding space and storage rack **35**, and between the unit holding space and operation jig **44**. At the time, since the height of unit conveyance robot **6** is appropriately maintained by the engagement of the guide member and the movement guide, the reception and delivery operation is stabilized. Unit conveyance robot **6** is controlled by set-up control section **8**.

2. Configuration Related to Control of Set-Up System **1**

(28) Next, a configuration related to control of set-up system **1** will be described. As illustrated in FIG. **3**, set-up control section **8** is communicatively connected to production management device **95**. Production management device **95** is communicatively connected to line control section **96**. Further, line control section **96** is communicatively connected to each of multiple types of board work machines constituting board work line **97**. Board work line **97** is a production line for mounting components on a board to mass-produce a board product. Component mounter **98** constituting board work line **97** is installed with set-up component supply unit **92** to enable the supply of the components.

(29) Production management device **95** manages a production plan and a production progress status of the board product. Line control section **96** controls the operation of board work line **97** based on the production plan received from production management device **95**. Further, line control section **96** sequentially transmits the production progress status in board work line **97** to production management device **95**.

(30) On the other hand, set-up control section **8** is communicatively connected to warehouse control section **29** of container warehouse **2** and storage shed control section **39** of unit storage shed **3**. Further, set-up control section **8** is connected to code reader **45** and generates load information V from the read result. Further, set-up control section **8** controls container conveyance robot **5** and unit conveyance robot **6**. Further, load information memory section **81** included in set-up control section **8** stores load information V. Set-up control section **8** includes a man-machine interface (an input section, a display section, a wireless communication section, or the like) for exchanging information with the operator.

(31) Various information including load information V is appropriately transmitted and received between the control elements described above. Further, all of each control element described above needs not to be independent hardware. For example, set-up control section **8**, warehouse control section **29**, and storage shed control section **39** may be achieved by different software of one computer device. Further, for example, set-up control section **8** may be achieved as a partial function of production management device **95**. Further, the communication connection or the information transmission described above may be performed by using a wireless communication device. The control functions of set-up control section **8** and the like will be described in detail in the following description of operations.

3. Operation of Set-Up System **1**

(32) Next, the operation of set-up system **1** will be described with reference to FIG. **4**. In step **S1** in FIG. **4**, set-up control section **8** determines whether a component replenishment timing due to a component shortage is close with reference to the production progress status of the board product managed by production management device **95**. For example, it is determined whether the remaining number of components of component container **91** of component supply unit **92** installed on component mounter **98** is decreased to a predefined number or less. When the component replenishment timing is close, set-up control section **8** determines that right now is the timing of performing the loading operation and executes steps **S2** to **S5**. When the component replenishment timing is not close, set-up control section **8** immediately advances the control to step **S11**. It is needless to say that the timing of performing the loading operation is determined with a margin exceeding the required time of the loading operation with respect to the component shortage prediction time.

(33) In step **S2**, set-up control section **8** notifies the operator that the loading operation is necessary. In the next step **S3**, set-up control section **8** selects component container **91** of a component type requiring replenishment. Further, set-up control section **8** designates the identification code of component container **91**, commands warehouse control section **29** to perform retrieval, and commands container conveyance robot **5** to perform conveyance. Warehouse control section **29** controls actuator **22** in accordance with the command and sends component container **91** to retrieval port **21**. Container conveyance robot **5** conveys component container **91** at retrieval port **21** to the vicinity of operation jig **44** in accordance with the command.

(34) In the next step **S4**, set-up control section **8** selects component supply unit **92** that can be loaded with component container **91**. Further, set-up control section **8** designates the identification code of component supply unit **92** and commands unit conveyance robot **6** to convey component supply unit **92**. Unit conveyance robot **6** picks up component supply unit **92** from unit storage shed **3** and conveys the same in accordance with the command, and causes operation jig **44** to hold the same. Steps **S2**, **S3**, and **S4** may be executed in parallel, or the execution procedure may be exchanged.

(35) In the next step **S5**, the operator who receives the notification moves component feed unit **92** to operation table **4** to perform the loading operation. Specifically, first, the operator receives component container **91** from container conveyance robot **5**. Secondly, the operator causes the code reader **45** to read the identification codes of component container **91** and component supply unit **92**. As a result, set-up control section **8** creates load information **V**. Created load information **V** is shared by unit storage shed **3** and component mounter **98** by the communication or the like.

(36) Third, the operator loads component supply unit **92** with component container **91**. The operator transports set-up component supply unit **92**, which is loaded with component container **91**, to component mounter **98** or instructs set-up control section **8** to temporarily store the same. Set-up control section **8** that has received the instruction causes unit conveyance robot **6** to convey and store set-up component supply unit **92** from operation jig **44** to unit storage shed **3**. After the end of step **S5**, set-up control section **8** advances the control to step **S11**.

(37) In step **S11**, set-up control section **8** determines whether the timing of performing set-up change for changing the type of the board product is close by referring to the production plan and the production progress status of the board product. For example, it is determined whether the remaining number of produced board products of the currently produced type is decreased to a predefined number or less. When the timing of performing the set-up change is close, set-up control section **8** determines that right now is the timing of performing the loading operation and executes steps **S12** to **S15**. When the timing of performing the set-up change is not close, set-up control section **8** immediately returns the control to step **S1**. It is needless to say that the timing of performing the loading operation is determined with a margin exceeding the required time for necessary multiple loading operations with respect to the estimated production completion time of the current board product.

(38) Since the operations of steps S12 to S15 are the same as the operations of steps S2 to S5, descriptions thereof will be omitted. In the set-up change, it is generally necessary to perform multiple loading operations to prepare for multiple component supply units 92. Therefore, in step S16, set-up control section 8 determines whether all component supply units 92 required for the set-up change are prepared. While all component feed units 92 are not prepared, set-up control section 8 and the operator repetitively executes steps S13 to S15. When all component supply units 92 are prepared by the repetition, set-up control section 8 returns the control to step S1.

(39) In component supply unit set-up system 1 of the first embodiment, component container 91 is conveyed from container warehouse 2 to operation table 4 by container conveyance robot 5, and component supply unit 92 is conveyed from unit storage shed 3 to operation table 4 by unit conveyance robot 6. Further, component supply unit 92, which is loaded with component container 91, on operation table 4 is returned to unit storage shed 3 by unit conveyance robot 6. Therefore, labor saving of related operations such as conveyance performed before and after the loading operation is achieved regardless of whether the loading operation is automated.

(40) Further, set-up control section 8 can timely determine the timing of performing the loading operation based on the production plan and the production progress status of the board product. According to this configuration, since component supply unit 92, in which the loading operation was ended and the set-up was already performed, does not increase unnecessarily, the labor for management is reduced. Further, component supply unit 92 does not become a work-in-process product that cannot be diverted to other uses, so that efficient operation is enabled. Further, it is possible to reduce stagnation for the mounting operation of component mounter 98 due to the delay of the loading operation. Further, by disposing the devices related to the loading operation adjacent to each other, a space-saving system configuration is achieved.

4. Component Supply Unit Set-Up System 1A of Second Embodiment

(41) Next, set-up system 1A of a second embodiment will be described with reference to FIG. 5 mainly on points different from the first embodiment. As illustrated in FIG. 5, operation table 4A of the second embodiment includes container loader 46. Container loader 46 automatically performs the loading operation performed by the operator in the first embodiment. The other configurations of the second embodiment are the same as those of the first embodiment.

(42) Container loader 46 is attached to an upper portion of base 41 in place of operation jig 44. Container loader 46 is formed in a box shape larger than operation jig 44. Container insertion port 47 through which component container 91 is inserted is provided on the upper face of container loader 46. Unit insertion port 48 through which component supply unit 92 is inserted is provided on the front face of container loader 46. First code reader 491 for reading the identification code of component container 91 and second code reader 492 for reading the identification code of component supply unit 92 are provided in container loader 46.

(43) Container conveyance robot 5 inserts component container 91 conveyed from container warehouse 2 into container insertion port 47. Further, unit conveyance robot 6 inserts component supply unit 92 conveyed from unit storage shed 3 into unit insertion port 48. When component container 91 and component supply unit 92 are inserted, first code reader 491 and second code reader 492 are operated, respectively. As a result, load information V is generated in container loader 46.

(44) Thereafter, container loader 46 automatically performs the loading operation. When the loading operation is normally ended, load information V is transmitted to set-up control section 8. Component supply unit 92 that is loaded with component container 91 is conveyed to unit storage shed 3 by unit conveyance robot 6. According to the second embodiment, automation of the loading operation and the related operations such as conveyance performed before and after the loading operation is achieved.

5. Component Supply Unit Set-Up System 1B of Third Embodiment

(45) Next, set-up system 1B of a third embodiment will be described with reference to FIGS. 6 and

7 mainly on points different from the first and second embodiments. In the third embodiment, landing station **61**, unit storage **64**, traveling path **65**, and conveyance vehicle **66** are added to the configuration of the first embodiment.

(46) As illustrated in FIG. **6**, landing station **61** is disposed side by side on the left of unit storage shed **3**. Landing station **61** also serves as a carry-out station for carrying out component supply unit **92** in which component container **91** is loaded toward the component mounter **98** and a carry-in station for carrying in component supply unit **92** that has been used in the component mounter **98**. At landing station **61**, the loading and unloading of conveyance vehicle **66** to be started and arrival are performed, and further, the set-up on unit storage **64** is performed. Landing station **61** is formed in a rectangular parallelepiped frame shape.

(47) Upper section guide **62** extending in the left-right direction is provided on an upper portion of the front face of landing station **61**. Lower section guide **63** extending in the left-right direction is provided below the front face of landing station **61**. Upper section guide **62** and lower section guide **63** are connected to upper section guide **32** and lower section guide **33** of unit storage shed **3** side by side to form a long movement guide. Therefore, unit conveyance robot **6** can move to the front side of landing station **61**.

(48) Two unit storages **64** are mounted on the upper face of landing station **61**. Unit storage **64** stores multiple component supply units **92**. Unit storage **64** is installed on component mounter **98** in a state in which multiple component supply units **92** are stored, so as to enable the supply of components. Unit storage **64** need not be installed in component mounter **98** but may improve the efficiency of the set-up work, the transportation, or the like of multiple component supply units **92**.

(49) As illustrated in FIG. **7**, traveling path **65** is laid from the rear side of landing station **61** to board work line **97**. Conveyance vehicle **66** travels on traveling path **65** in accordance with a wireless command from set-up control section **8** and also loads and unloads unit storage **64**. Conveyance vehicle **66** appropriately reports a progress status of the conveyance operation to set-up control section **8**. The standby position of conveyance vehicle **66** is set on the rear side of landing station **61**. Traveling path **65** may be laid over multiple board work lines **97** so that multiple conveyance vehicles **66** may travel while avoiding collisions. Further, conveyance vehicle **66** may be an AGV of a type in which physical traveling path **65** is unnecessary and travels by referring to information about the traveling path on map data.

(50) In the above configuration, the operator performs the loading operation on operation table **4** as described in the first embodiment. Next, unit conveyance robot **6** conveys component supply unit **92**, which is loaded with component container **91**, from operation table **4** to landing station **61** and stores component supply unit **92** in unit storage **64**. The operation of the operator and the operation of unit conveyance robot **6** described above are repeated so that unit storage **64** is ready to be used.

(51) Thereafter, conveyance vehicle **66** removes unit storage **64** ready to be used from landing station **61** and loads the same. Next, conveyance vehicle **66** travels to component mounter **98**, conveys unit storage **64**, and installs unit storage **64** in component mounter **98**. Alternatively, conveyance vehicle **66** delivers unit storage **64** to the operator in the vicinity of component mounter **98**. According to the third embodiment, in addition to the related operations before and after the loading operation, automation of the operation of storing multiple component supply units **92** in unit storage **64** and the operation of transporting unit storage **64** to component mounter **98** is achieved.

6. Application and Modification of Embodiment

(52) The determination of the component replenishment timing and the timing of performing the set-up change described in the operation flow of FIG. **4** may be performed by production management device **95**. In this case, set-up control section **8** advances the control according to the command of performing the loading operation from production management device **95**. Further, multiple unit storage sheds **3** may be provided to separately store component supply unit **92**, which is loaded with component container **91**, and component supply unit **92**, which is not loaded with

component container **91**. Further, in configurations in which component mounter **98** are disposed close to set-up systems (**1**, **1A**), unit conveyance robot **6** may convey component supply unit **92**, which is loaded with component container **91**, to component mounter **98**.

(53) Further, in the third embodiment, container loader **46** of the second embodiment may be applied in place of operation jig **44**. In the third embodiment, conveyance vehicle **66** may return unit storage **64** that has been used in component mounter **98** to landing station **61**. Thereafter, component container **91** is removed from component supply unit **92** on operation table **4**.

Component supply unit **92** is stored in unit storage shed **3**, and component container **91** is returned and stored to and in container warehouse **2**. Further, the first to third embodiments can be applied or modified in various manners.

REFERENCE SIGNS LIST

(54) **1**, **1A**, **1B**: component supply unit set-up system **2**: container warehouse **21**: retrieval port **22**: actuator **23**: code reader **29**: warehouse control section **3**: unit storage shed **32**: upper section guide **33**: lower section guide **35**: storage rack **39**: storage shed control section **4**, **4A**: operation table **42**: upper section guide **43**: lower section guide **44**: operation jig **45**: code reader **46**: container loader **5**: container conveyance robot **53**: arm portion **6**: unit conveyance robot **61**: landing station **62**: upper section guide **63**: lower section guide **64**: unit storage **66**: conveyance vehicle **8**: set-up control section **81**: load information memory section **91**: component container **92**: component supply unit **95**: production management device **98**: component mounter **V**: load information

Claims

1. A component supply unit set-up system comprising: a container warehouse configured to store multiple component containers that contain multiple components; a unit storage shed configured to store multiple component supply units that supply the components, which are contained in the component containers, in a mounting operation performed by a component mounter; an operation table disposed between the container warehouse and the unit storage shed, the operation table including a container loader that automatically performs a loading operation to load the component supply unit with the component container; a container conveyance robot configured to convey the component container between the container warehouse and the operation table; and a unit conveyance robot configured to convey the component supply unit between the unit storage shed and the operation table, wherein the container loader automatically loads the component supply unit conveyed by the unit conveyance robot with the component container conveyed by the container conveyance robot.

2. The component supply unit set-up system according to claim 1, wherein the container conveyance robot is a non-moving robot that does not need to move between the container warehouse and the operation table disposed close to each other.

3. The component supply unit set-up system according to claim 1, further comprising: a movement guide provided from the unit storage shed to the operation table, wherein the unit conveyance robot is a moving robot that moves along the movement guide.

4. The component supply unit set-up system according to claim 3, wherein the movement guide is configured such that section guides provided in each of the unit storage shed and the operation table are lined up in a movement direction of the unit conveyance robot.

5. The component supply unit set-up system according to claim 4, wherein the unit storage shed is capable of being additionally installed.

6. The component supply unit set-up system according to claim 1, wherein the component container is a tape reel in which a carrier tape containing the multiple components is wound, and the component supply unit is a feeder device loaded with the tape reel or is a reel cassette loaded with the tape reel and formed separately from a feeder main body portion having a feeding mechanism of the carrier tape.

7. A component supply unit set-up system comprising: a container warehouse configured to store multiple component containers for containing multiple components; a unit storage shed configured to store multiple component supply units used when the components, which are contained in the component containers, are supplied in a mounting operation performed by a component mounter; an operation table disposed between the container warehouse and the unit storage shed and used for a loading operation to load the component supply unit with the component container; a container conveyance robot configured to convey the component container between the container warehouse and the operation table; a unit conveyance robot configured to convey the component supply unit between the unit storage shed and the operation table; and a set-up control section configured to control for selecting the component container to be conveyed from the container warehouse to the operation table, and determine a timing of performing the loading operation of the selected and conveyed component container on the operation table, based on a command from a production management device that manages at least one of a production plan and a production progress status of a board product on which the component is mounted or with reference to at least one of the production plan and the production progress status.

8. The component supply unit set-up system according to claim 7, further comprising: a load information memory section configured to store load information in which an individual of the component containers and an individual of the component supply units, which are loaded with the component containers, are associated with each other.
